

Prevalence of premenstrual dysphoric disorder among school-going adolescent girls.

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Premenstrual dysphoric disorder (PMDD) is a severe form of premenstrual syndrome which has a significant negative impact on the various domains of life of adolescent girls. The aim of this study was to estimate the prevalence of PMDD in adolescent girls studying in classes 7th-10th and ascertain the level of stress, anxiety, and depression among them. A cross-sectional study was conducted in three all-girls schools in Chandigarh (two - government and one - private) after taking necessary permissions. Participants were evaluated on a self-rated questionnaire which included the PMDD scale, Perceived Stress Scale, Generalized Anxiety Disorder-7 (GAD-7) questionnaire, and Patient Health Questionnaire-9. A total of 397 girls participated in the study. The mean age of respondents was 14.34 (standard deviation [SD]: 1.17; range: 11-20) years, with most (44%) studying in class 10th. The mean age of attaining menarche was 12.54 (SD: 0.92; range: 10-15) years. The prevalence of PMDD was found to be 4.8% (n = 19). Majority of the respondents reported moderate levels of perceived stress (62%). A positive correlation was seen between the severity of PMDD, age, levels of perceived stress, severity of depression, and anxiety in the respondents. Nearly 5% of adolescent girls suffer from PMDD, with higher prevalence among those with depression, GAD, and higher perceived stress. Thus, there is a need to screen adolescent females for PMDD at the earliest and institute intervention to minimize its negative impact.

The prevalence of premenstrual dysphoric disorder in a randomly selected group of urban and rural women.

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Premenstrual dysphoric disorder (PMDD) was included as a provisional diagnostic category in the appendices of Diagnostic and Statistical Manual of Mental Disorders (DSM)-III-R (then called late luteal phase dysphoric disorder) and remained as an appendix in DSM-IV. Our study aimed to determine the prevalence of PMDD using all four DSM-IV research diagnostic criteria in a representative sample of women of reproductive age in the United States. Data were collected in the homes of women between the ages of 13 and 55 years in two urban and two rural sites using a random sampling procedure developed by the National Opinion Research Center. Women completed daily symptom questionnaires and provided urine specimens each day for two consecutive ovulatory menstrual cycles (ovulation was estimated for women taking oral contraceptives) and were screened for psychiatric disorders by trained interviewers. Symptoms were counted toward a diagnosis of PMDD if they worsened significantly during the late luteal week during two consecutive ovulatory menstrual cycles, occurred on days in which women reported marked interference with functioning, and were not due to another mental disorder. In the final analysis, 1246 women who had had at least one menstrual cycle and were neither naturally nor surgically menopausal nor pregnant were selected. Of the women in the study, 1.3% met criteria for the diagnosis as defined in DSM-IV. The prevalence of PMDD is considerably lower than DSM-IV estimates and all but one of the estimates obtained from previous studies when all DSM-IV diagnostic criteria are considered. We suggest a new process for diagnosing PMDD based on our findings.

How does premenstrual dysphoric disorder relate to depression and anxiety disorders?

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Premenstrual dysphoric disorder (PMDD) is a severe variant of premenstrual syndrome that afflicts approximately 5% of all women of fertile age. The hallmark of this condition is the surfacing of symptoms during the luteal phase of the menstrual cycle, and the disappearance of symptoms shortly after the onset of menstruation. Whereas many researchers have emphasized the similarities between PMDD and anxiety disorders, and in particular panic disorder, others have suggested that PMDD should be regarded as a variant of depression. Supporting both these notions, the treatment of choice for PMDD, the serotonin reuptake inhibitors (SRIs), is also first line of treatment for depression and for most anxiety disorders. In this review, the relationship between PMDD on the one hand, and anxiety and depression on the other, is being discussed. Our conclusion is that PMDD is neither a variant of depression nor an anxiety disorder, but a distinct diagnostic entity, with irritability and affect lability rather than depressed mood or anxiety as most characteristic features. The clinical profile of SRIs when used for PMDD, including a short onset of action, suggests that this effect is mediated by other serotonergic synapses than the antidepressant and anti-anxiety effects of these drugs. Although we hence suggest that PMDD should be regarded as a distinct entity, it should be emphasized that this disorder does display intriguing similarities with other conditions, and in particular with panic disorder, which should be the subject of further studies. Also, the possibility that there are subtypes of PMDD more closely related to depression, or anxiety disorders, than the most common form of the syndrome, should not be excluded.

GABAergic neuroactive steroid response to sertraline in premenstrual dysphoric disorder.

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Premenstrual dysphoric disorder (PMDD) affects approximately 5% of menstruating individuals, with significant negative mood symptoms in the luteal phase of the menstrual cycle. PMDD's pathophysiology and treatment mechanisms are poorly characterized, but may involve altered neuroactive steroid function in the brain. Selective serotonin reuptake inhibitors (SSRIs), a first-line PMDD treatment, reportedly alter gamma-aminobutyric acid (GABA)ergic neuroactive steroid levels in PMDD. The aims of this study were to determine whether the SSRI sertraline increased serum levels of neuroactive steroids that modulate the effect of GABA at GABA-A receptors (GABAAR) and if so, whether an increase was associated with improvement in PMDD symptoms. Participants included controls and individuals with PMDD. Serum levels of 9 neuroactive steroids were measured (3 α ,5 α -THP; 3 α 5 β -THP; pregnenolone; 3 α ,5 α -androsterone; 3 α ,5 β -androsterone; 3 α ,5 α -A-diol; 3 α 5 β -A-diol; 3 α ,5 α -THDOC; 3 α 5 β -THDOC) in the follicular and luteal phases. In the subsequent luteal phase, neuroactive steroids were measured during sertraline treatment (50 mg sertraline from approximate ovulation to menses onset) in the PMDD group. Mixed models assessed associations among diagnostic group, menstrual cycle phase, and sertraline treatment. Participants included 38 controls and 32 women with PMDD. There were no significant differences in neuroactive steroid levels between controls and participants with PMDD in the luteal phase ($p > 0.05$). Within the PMDD group, sertraline treatment significantly increased serum pregnanolone levels and the pregnanolone:progesterone ratio, and decreased 3 α ,5 α -androsterone. This was the first study to assess the impact of SSRI treatment on peripheral levels of GABAergic neuroactive steroids in PMDD. Within the PMDD group, sertraline treatment was associated with a significant increase in luteal phase serum pregnanolone levels and a significantly increased pregnanolone:progesterone ratio, a novel finding. Future research should examine alterations in the metabolic pathways among GABAergic neuroactive steroids in individuals with PMDD, in a placebo-controlled design.

Estrogen receptor alpha (ESR-1) associations with psychological traits in women with PMDD and controls.

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Premenstrual Dysphoric Disorder (PMDD) is a mood disorder affecting about 5% of women and is associated with substantial morbidity. Albeit inconsistently, PMDD is described as being characterized by heritable personality traits. Although PMDD is a heritable disorder, it is unclear whether any of the heritable susceptibility to PMDD resides in heritable personality traits. In groups of carefully characterized women with PMDD (n=68) and controls (n=56), we attempted to determine whether diagnosis-related traits could be confirmed, as well as to determine whether such traits were associated with SNPs in estrogen receptor alpha (ESR-1) that we previously demonstrated were associated with PMDD. We observed 7/25 traits to be significantly different in patients and controls and further showed that 11/12 significant associations observed between these 7 traits and 16 ESR-1 SNPs involved the intron 4 SNPs previously shown to be the locus of the association with PMDD. While several interactions between genotype and diagnosis were observed, the effect of genotype in most instances was in the same direction in patients and controls. These data demonstrate affective state-independent personality traits that distinguish patients with PMDD from controls and further support the relevance of ESR-1 polymorphic variants in the regulation of non-reproductive behaviors.

Copy number variation profile-based genomic typing of premenstrual dysphoric disorder in Chinese.

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Premenstrual dysphoric disorder (PMDD) affects nearly 5% of women of reproductive age. Symptomatic heterogeneity, together with largely unknown genetics, has greatly hindered its effective treatment. In the present study, analysis of genomic sequencing-based copy number variations (CNVs) called from 100 kb white blood cell DNA sequence windows by means of semisupervised clustering led to the segregation of patient genomes into the D and V groups, which correlated with the depression and invasion clinical types, respectively, with 89.0% consistency. Application of diagnostic CNV features selected using the correlation-based machine learning method enabled the classification of the CNVs obtained into the D group, V group, total patient group, and control group with an average accuracy of 83.0%. The power of the diagnostic CNV features was 0.98 on average, suggesting that these CNV features could be used for the molecular diagnosis of the major clinical types of PMDD. This demonstrated concordance between the CNV profiles and clinical types of PMDD supported the validity of symptom-based diagnosis of PMDD for differentiating between its two major clinical types, as well as the predominantly genetic nature of PMDD with a host of overlaps between multiple susceptibility genes/pathways and the diagnostic CNV features as indicators of involvement in PMDD etiology.

An ER-resident membrane protein complex regulates nicotinic acetylcholine receptor subunit composition at the synapse.

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Nicotinic acetylcholine receptors (nAChRs) are homo- or heteropentameric ligand-gated ion channels mediating excitatory neurotransmission and muscle activation. Regulation of nAChR subunit assembly and transfer of correctly assembled pentamers to the cell surface is only partially understood. Here, we characterize an ER transmembrane (TM) protein complex that influences nAChR cell-surface expression and functional properties in *Caenorhabditis elegans* muscle. Loss of either type I TM protein, NRA-2 or NRA-4 (nicotinic receptor associated), affects two different types of muscle nAChRs and causes in vivo resistance to cholinergic agonists. Sensitivity to subtype-specific agonists of these nAChRs is altered differently, as demonstrated by whole-cell voltage-clamp of dissected adult muscle, when applying exogenous agonists or after photo-evoked, channelrhodopsin-2 (ChR2) mediated acetylcholine (ACh) release, as well as in single-channel recordings in cultured embryonic muscle. These data suggest that nAChRs desensitize faster in *nra-2* mutants. Cell-surface expression of different subunits of the 'levamisole-sensitive' nAChR (L-AChR) is differentially affected in the absence of NRA-2 or NRA-4, suggesting that they control nAChR subunit composition or allow only certain receptor assemblies to leave the ER.

[Molecular mechanisms underlying the formation of neuromuscular junction].

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The neuromuscular junction (NMJ) is a synapse between a motor neuron and skeletal muscle. The contraction of skeletal muscle is controlled by the neurotransmitter acetylcholine (ACh), which is released from the motor nerve terminal. To achieve efficient neuromuscular transmission, acetylcholine receptors (AChRs) must be densely clustered on the muscle membrane of the NMJ. Failure of AChR clustering is associated with disorders of neuromuscular transmission such as congenital myasthenic syndromes (CMS) and myasthenia gravis (MG). Motoneuronal agrin and muscle-specific receptor tyrosine kinase (MuSK) are known to play essential roles in the formation and maintenance of NMJs in the central region of each muscle. However, it had been unclear how agrin activates MuSK. Recent studies have elucidated the roles of several key molecules, including the cytoplasmic adaptor protein Dok-7 and LDL receptor-related protein 4 (Lrp4), in agrin-induced MuSK activation. Moreover, new evidence indicates that cyclin-dependent kinase 5 (Cdk5) regulates postsynaptic differentiation. In this review, we summarize the latest developments in molecular mechanisms underlying NMJ formation in vertebrates.

Synaptic retrograde regulation of the PKA-induced SNAP-25 and Synapsin-1 phosphorylation.

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Bidirectional communication between presynaptic and postsynaptic components contribute to the homeostasis of the synapse. In the neuromuscular synapse, the arrival of the nerve impulse at the presynaptic terminal triggers the molecular mechanisms associated with ACh release, which can be retrogradely regulated by the resulting muscle contraction. This retrograde regulation, however, has been poorly studied. At the neuromuscular junction (NMJ), protein kinase A (PKA) enhances neurotransmitter release, and the phosphorylation of the molecules of the release machinery including synaptosomal associated protein of 25 kDa (SNAP-25) and Synapsin-1 could be involved. Accordingly, to study the effect of synaptic retrograde regulation of the PKA subunits and its activity, we stimulated the

rat phrenic nerve (1 Hz, 30 min) resulting or not in contraction (abolished by μ -conotoxin GIIIB). Changes in protein levels and phosphorylation were detected by western blotting and cytosol/membrane translocation by subcellular fractionation. Synapsin-1 was localized in the levator auris longus (LAL) muscle by immunohistochemistry. Here we show that synaptic PKA C β subunit regulated by RII β or RII α subunits controls activity-dependent phosphorylation of SNAP-25 and Synapsin-1, respectively. Muscle contraction retrogradely downregulates presynaptic activity-induced pSynapsin-1 S9 while that enhances pSNAP-25 T138. Both actions could coordinately contribute to decreasing the neurotransmitter release at the NMJ. This provides a molecular mechanism of the bidirectional communication between nerve terminals and muscle cells to balance the accurate process of ACh release, which could be important to characterize molecules as a therapy for neuromuscular diseases in which neuromuscular crosstalk is impaired.

Glutamate at the Vertebrate Neuromuscular Junction: From Modulation to Neurotransmission.

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Although acetylcholine is the major neurotransmitter operating at the skeletal neuromuscular junction of many invertebrates and of vertebrates, glutamate participates in modulating cholinergic transmission and plastic changes in the last. Presynaptic terminals of neuromuscular junctions contain and release glutamate that contribute to the regulation of synaptic neurotransmission through its interaction with pre- and post-synaptic receptors activating downstream signaling pathways that tune synaptic efficacy and plasticity. During vertebrate development, the chemical nature of the neurotransmitter at the vertebrate neuromuscular junction can be experimentally shifted from acetylcholine to other mediators (including glutamate) through the modulation of calcium dynamics in motoneurons and, when the neurotransmitter changes, the muscle fiber expresses and assembles new receptors to match the nature of the new mediator. Finally, in adult rodents, by diverting descending spinal glutamatergic axons to a denervated muscle, a functional reinnervation can be achieved with the formation of new neuromuscular junctions that use glutamate as neurotransmitter and express ionotropic glutamate receptors and other markers of central glutamatergic synapses. Here, we summarize the past and recent experimental evidences in support of a role of glutamate as a mediator at the synapse between the motor nerve ending and the skeletal muscle fiber, focusing on the molecules and signaling pathways that are present and activated by glutamate at the vertebrate neuromuscular junction.